

A 12-vertex planar graph with $\text{fvs} = 5$ and $\text{fp} = 2$: a counterexample to the face-packing conjecture of Bonamy, Dross, Masařík, Nadara, Pilipczuk and Pilipczuk

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Abstract

A *face-packing* of a planar graph G is a set of vertex-disjoint cycles that, for some planar embedding of G , all bound faces; $\text{fp}(G)$ is the largest size of one. Bonamy, Dross, Masařík, Nadara, Pilipczuk and Pilipczuk conjectured (Conjecture 2 of arXiv:1912.01570, Conjecture 14 of the published version) that every planar graph satisfies $\text{fvs}(G) \leq 2 \text{fp}(G)$, a strengthening of Jones' Conjecture $\text{fvs}(G) \leq 2 \text{cp}(G)$. We show that this is false: the 12-vertex planar graph obtained from a 4-cycle $abcd$ by attaching a K_4 at a and c and another K_4 at b and d has $\text{fvs} = 5$ and $\text{fp} = 2$. More generally, a family $(G_t)_{t \geq 1}$ of planar graphs with $\text{fvs}(G_t) = 3t + 2$ and $\text{fp}(G_t) = t + 1$ shows that no inequality $\text{fvs} \leq c \text{fp}$ holds for planar graphs with a constant $c < 3$. The family has $\text{cp}(G_t) = 2t + 1$ and so does not touch Jones' Conjecture itself.

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and it has not been checked by a human mathematician. [Section 4](#) records the full provenance.

1 Introduction

All graphs are finite and simple. A *cycle packing* of a graph G is a set of pairwise vertex-disjoint cycles of G , and $\text{cp}(G)$ denotes the largest size of a cycle packing. A *feedback vertex set* is a set $S \subseteq V(G)$ such that $G - S$ is a forest, and $\text{fvs}(G)$ is the least size of one. Trivially $\text{fvs}(G) \geq \text{cp}(G)$. Jones' Conjecture, first recorded by Kloks, Lee and Liu [4], asserts that $\text{fvs}(G) \leq 2 \text{cp}(G)$ for every planar graph G ; the factor 2 would be best possible because of K_4 (or of any wheel). The best general bound is $\text{fvs}(G) \leq 3 \text{cp}(G)$, and Bonamy, Dross, Masařík, Nadara, Pilipczuk and Pilipczuk [1, 2] proved Jones' Conjecture for subcubic planar graphs.

In the concluding paragraph of that paper the authors pose a stronger conjecture, in which the packed cycles are required to be faces. Following them, a *face-packing* of a planar graph G is a cycle packing such that, for some planar embedding of G , every cycle in the packing bounds a face of that embedding; $\text{fp}(G)$ is the largest size of a face-packing. Since every face-packing is a cycle packing, $\text{fp}(G) \leq \text{cp}(G)$. The statement, quoted verbatim from [1], is:

Conjecture 1.1 ([1, Conjecture 2]; [2, Conjecture 14]). For every planar graph G , we have $\text{fvs}(G) \leq 2 \cdot \text{fp}(G)$, where $\text{fp}(G)$ is the maximum size of a face-packing of G , i.e., a cycle-packing where, for some embedding of G , every cycle bounds a face.

The authors motivate the conjecture as a way “to highlight how little we understand around Jones' Conjecture”. They note that the example of many nested disjoint cycles shows that the embedding cannot be fixed in advance (so the quantifier “for some embedding” is essential), and that the discharging argument giving $\text{fvs}(G) \leq 3 \text{cp}(G)$ does not yield [Conjecture 1.1](#) even with the factor 3 in place of 2. The conjecture is stated word for word identically, and with the same quantifier “for every planar

graph G'' , in the arXiv version [1] and in the published version [2]; neither imposes any connectivity, degree or further hypothesis on G .

The only subsequent work on this circle of questions that we are aware of is by Bärnkopf and Györi [3], who prove Jones' Conjecture for Halin graphs and for the class of *based planar graphs* (plane graphs having a face adjacent to every other face). Their Remark 2.5 observes that every based planar graph even satisfies $\text{fvs}(G) \leq 2 \text{fp}(G)$. Neither that paper nor either of the other two works citing [1] that the referee located (Remark 3.5) refutes Conjecture 1.1.

We show that Conjecture 1.1 is false, already on 12 vertices, and that the factor 2 cannot be replaced by any constant below 3.

The counterexample

Let G_1 be the graph with 12 vertices $a, b, c, d, p, q, r, s, p', q', r', s'$ and the following 20 edges: the 4-cycle $abcd$ (edges ab, bc, cd, da); all six edges among $\{p, q, r, s\}$ and all six among $\{p', q', r', s'\}$ (two copies of K_4); and the four attachment edges ap, cq, bp', dq' . In words: a 4-cycle with a K_4 hung on the pair of opposite vertices a, c and another K_4 hung on the other pair b, d , using two distinct vertices of each K_4 . Every vertex has degree 3 except p, q, p', q' , which have degree 4.

Theorem 1.2. G_1 is a planar graph with $\text{fvs}(G_1) = 5$ and $\text{fp}(G_1) = 2$. In particular $\text{fvs}(G_1) = 5 > 4 = 2 \text{fp}(G_1)$, so Conjecture 1.1 is false.

G_1 is the case $t = 1$ of the following family.

Definition 1.3 (the graphs G_t). Let $t \geq 1$. Take pairwise vertex-disjoint 4-cycles $C_i = a_i b_i c_i d_i a_i$ ($1 \leq i \leq t$) and pairwise vertex-disjoint copies Q_j of K_4 on vertex sets $\{p_j, q_j, r_j, s_j\}$ ($0 \leq j \leq t$), all these vertex sets being mutually disjoint, and add exactly the following edges:

- $a_i p_i$ and $c_i q_i$ for $1 \leq i \leq t$ (the gadget Q_i is attached to C_i at its opposite vertices a_i, c_i);
- $d_i b_{i+1}$ for $1 \leq i < t$ (the cycles form a chain);
- $b_1 p_0$ and $d_t q_0$ (the gadget Q_0 closes the chain).

The edges $a_i p_i, c_i q_i, b_1 p_0, d_t q_0$ are called *attachment edges*; each gadget Q_j is incident with exactly two of them, one at p_j and one at q_j .

For $t = 1$ the vertices $a_1, b_1, c_1, d_1, Q_1, Q_0$ are $a, b, c, d, \{p, q, r, s\}, \{p', q', r', s'\}$ above. In general G_t has $4t + 4(t + 1) = 8t + 4$ vertices and $4t + 6(t + 1) + 2t + (t - 1) + 2 = 13t + 7$ edges; every vertex has degree 3 except the $2t + 2$ vertices p_j, q_j , which have degree 4.

Theorem 1.4 (Main theorem). For every integer $t \geq 1$ the graph G_t is a simple planar graph of order $8t + 4$ and maximum degree 4 with

$$\text{fvs}(G_t) = 3t + 2, \quad \text{fp}(G_t) = t + 1, \quad \text{cp}(G_t) = 2t + 1.$$

Corollary 1.5. Conjecture 1.1 is false: $\text{fvs}(G_1) = 5 > 4 = 2 \text{fp}(G_1)$. Moreover, for every constant $c < 3$ there is a planar graph G with $\text{fvs}(G) > c \text{fp}(G)$, since $\text{fvs}(G_t)/\text{fp}(G_t) = 3 - \frac{1}{t+1} \rightarrow 3$. On the other hand $\text{fvs}(G_t) = 3t + 2 \leq 4t + 2 = 2 \text{cp}(G_t)$, so the family is consistent with Jones' Conjecture.

Proof. Immediate from Theorem 1.4: $(3t + 2)/(t + 1) = 3 - 1/(t + 1)$, which exceeds any $c < 3$ for t large enough. $\square$

Idea of proof. Four elementary facts about G_t do all the work; the last one is the only place where planarity enters. (i) *Feedback vertex number.* Each C_i needs one deleted vertex and each $Q_j \cong K_4$ needs two, and these $2t + 1$ subgraphs are vertex-disjoint, so $\text{fvs}(G_t) \geq 3t + 2$; deleting $\{a_i\}_i \cup \{p_j, q_j\}_j$ leaves the path $b_1 c_1 d_1 b_2 c_2 d_2 \cdots b_t c_t d_t$ plus $t + 1$ isolated edges $r_j s_j$. (ii) *One packed cycle per gadget.* A cycle meeting Q_j either lies inside Q_j and uses at least three of its four vertices, or leaves Q_j and must

then use both attachment edges, hence both p_j and q_j ; any two such cycles intersect. (iii) *Gadget-free cycles.* After deleting all gadgets, the chain edges $d_i b_{i+1}$ are bridges, so the only remaining cycles are the C_i . (iv) *No C_i is ever a face.* The path $a_i p_i q_i c_i$ through Q_i and a $b_i - d_i$ path through the rest of the chain and Q_0 are internally disjoint from each other and from C_i , and their endpoints alternate on C_i . In any planar embedding the two paths lie on opposite sides of the Jordan curve C_i , so neither side is a face. From (ii)–(iv), in every planar embedding every facial cycle meets a gadget and distinct disjoint facial cycles meet distinct gadgets, so $\text{fp}(G_t) \leq t + 1$; a suitable drawing exhibits $t + 1$ disjoint facial triangles $p_j q_j s_j$. From (ii) and (iii), $\text{cp}(G_t) \leq (t + 1) + t$, attained by the C_i together with one triangle from each Q_j . [\[full proof\]](#)

2 Lemmas

We write $V_j = \{p_j, q_j, r_j, s_j\}$ for the vertex set of the gadget Q_j .

Lemma 2.1 (Planarity and a face-packing of size $t + 1$). *G_t is planar, and it has a planar embedding in which the $t + 1$ pairwise vertex-disjoint triangles $p_j q_j s_j$ ($0 \leq j \leq t$) all bound faces. Hence $\text{fp}(G_t) \geq t + 1$.*

Idea of proof. Draw the C_i as diamonds in a row, with b_i and d_i as the left and right corners, add the chain edges between consecutive diamonds, a vertical chord $a_i c_i$ inside each diamond and a closing edge $d_t b_1$ around the outside. Then replace each chord $a_i c_i$ by the gadget Q_i and the closing edge by Q_0 : since p_j and q_j are adjacent in K_4 they lie on a common face of the plane K_4 , so the gadget can be inserted in a small disc around a point of the edge it replaces. The three inner triangular faces of each plane K_4 , among them $p_j q_j s_j$, survive. [\[full proof\]](#)

Lemma 2.2 (Feedback vertex number). $\text{fvs}(G_t) = 3t + 2$.

Idea of proof. Lower bound from the $2t + 1$ vertex-disjoint subgraphs C_i ($\text{fvs} = 1$) and $Q_j \cong K_4$ ($\text{fvs} = 2$, as any three vertices of K_4 span a triangle). Upper bound from the explicit set $S = \{a_i : 1 \leq i \leq t\} \cup \{p_j, q_j : 0 \leq j \leq t\}$. [\[full proof\]](#)

Lemma 2.3 (One cycle per gadget). *Fix $0 \leq j \leq t$. Any two cycles of G_t that both contain a vertex of V_j have a common vertex. Consequently, in any cycle packing of G_t at most one cycle meets V_j .*

Idea of proof. A cycle inside Q_j has length 3 or 4 and so contains at least three of the four vertices of V_j . A cycle not inside Q_j that meets V_j crosses the edge cut between V_j and the rest an even, positive number of times; that cut consists of the two attachment edges of Q_j , so the cycle contains both p_j and q_j . Any two 3-subsets of V_j , any two supersets of $\{p_j, q_j\}$, and any 3-subset and any superset of $\{p_j, q_j\}$ intersect. [\[full proof\]](#)

Lemma 2.4 (Gadget-free cycles). *The cycles of G_t that contain no vertex of $V_0 \cup \dots \cup V_t$ are exactly $C_1, \dots, C_t$.*

Idea of proof. $G_t - \bigcup_j V_j$ consists of the cycles C_i joined into a chain by the edges $d_i b_{i+1}$, which are bridges of that graph. A cycle uses no bridge, so it lies within a single C_i . [\[full proof\]](#)

The next lemma isolates the topological input. Recall that a face of a plane graph is a connected component of the complement of the drawing, and that a cycle C bounds a face F if the frontier of F is exactly (the drawing of) C .

Lemma 2.5 (Alternating paths block a face). *Let H be a plane graph containing a cycle C and two paths P (from x to y) and R (from u to v), each with at least one edge, such that P and R are internally disjoint from each other and from C , and x, u, y, v are four distinct vertices appearing in this cyclic order on C . Then no face of H is bounded by C .*

Idea of proof. By the Jordan curve theorem $\mathbb{R}^2 \setminus C$ has two components (the two *sides* of C). The interior of P is connected and disjoint from C , so it lies in one side, S say. $C \cup P$ is a plane theta graph with three faces, bounded by C , by $xCu y Px$ and by $yCv x Py$ (the two cycles through P); the two latter faces partition $S \setminus P$, and u lies on the frontier of only one of them while v lies on the frontier of only the other. The interior of R is connected and disjoint from $C \cup P$, hence inside a single face of $C \cup P$ whose frontier contains both u and v : this is the face bounded by C , i.e. the side of C other than S . Thus both sides of C contain edges of H . A face bounded by C would be an open connected subset of one side with no frontier points inside that side, hence the whole side, which is impossible. [\[full proof\]](#)

Lemma 2.6 (No C_i is facial). *For every $1 \leq i \leq t$ and every planar embedding of G_t , the cycle C_i does not bound a face.*

Idea of proof. Apply [Lemma 2.5](#) with $C = C_i$, $P = P_i = a_i p_i q_i c_i$ and $R = R_i$ the b_i - d_i path that runs from b_i down the chain to b_1 (through $d_k c_k b_k$ for $k = i-1, \dots, 1$), crosses $b_1 p_0 q_0 d_t$, and returns from d_t up the chain to d_i (through $b_k c_k d_k$ for $k = t, \dots, i+1$). The endpoints a_i, b_i, c_i, d_i alternate on C_i , and P_i and R_i are internally disjoint from each other and from C_i . (Equivalently: $C_i \cup P_i \cup R_i$ is a subdivision of K_4 in which C_i corresponds to a 4-cycle, and 4-cycles of K_4 are not facial.) [\[full proof\]](#)

3 Remarks

Remark 3.1 (Scope of the statement). [Conjecture 1.1](#) is quantified over all planar graphs, in both [\[1\]](#) and [\[2\]](#), and the referee ([Appendix B](#)) checked both texts for silently assumed hypotheses and found none. G_1 is simple, connected and 2-connected, with maximum degree 4, so it is squarely within scope. The definition of fp used here is the one of [\[1\]](#): vertex-disjoint cycles, all bounding faces of one common embedding, maximised over embeddings. This is the reading most favourable to the conjecture; under the fixed-embedding reading of [\[3, Definition 1.5\]](#) fp can only decrease, so the refutation holds a fortiori. It even survives the reading in which each packed cycle may bound a face in an embedding of its own: the referee computed that exactly 20 cycles of G_1 bound a face in *some* planar embedding (none of them is C_1), and the largest pairwise vertex-disjoint subfamily of these 20 cycles still has size 2. Only an edge-disjoint reading of “packing” would rescue the statement, and that reading is excluded by the source paper’s definition of a cycle packing as a set of vertex-disjoint cycles.

Remark 3.2 (Independent computational verification). The referee recomputed both invariants of G_1 without using any of the arguments above (scripts in `verification_astra/scripts/1912.01570__00/`). $\text{fvs}(G_1) = 5$ was obtained by exhaustive search over all vertex subsets, with witness $\{a, p, q, p', q'\}$; $\text{fvs}(G_2) = 8$ likewise. $\text{fp}(G_1) = 2$ was obtained by enumerating all $6^4 \cdot 2^8 = 331\,776$ rotation systems of G_1 , keeping the 32 with $V - E + F = 2$ (the sphere embeddings), tracing their faces and maximising the number of pairwise vertex-disjoint facial cycles over each; the witnessing packing is $\{p, r, s\}$, $\{p', q', r'\}$, and C_1 is facial in none of the 32 embeddings. The rotation-system enumerator was validated on K_4 , the cube and the octahedron (2 sphere embeddings each, a mirror pair, as Whitney’s theorem [\[5\]](#) predicts for 3-connected planar graphs), on C_5 (1) and on K_5 and $K_{3,3}$ (0). A second, structurally independent upper bound was obtained from the *apex criterion*: if a cycle C bounds a face in some embedding of G then G plus a new vertex adjacent to all of $V(C)$ is planar, so $\text{fp}(G)$ is at most the largest disjoint family among the cycles passing this planarity test. On G_1 the apex test selects exactly the same 20 cycles that the exhaustive enumeration found to be facial-realisable, and it gives $\text{fp}(G_t) \leq t + 1$ for $t = 1, 2, 3$ (20, 36 and 58 candidate cycles respectively; no C_i passes). Together with the explicit embeddings of [Lemma 2.1](#), whose face lists the referee traced (Euler’s formula checked) for $t \leq 6$, this confirms $\text{fp}(G_t) = t + 1$ exactly for $t = 1, 2, 3$. The referee also rebuilt G_1 a second time from the one-sentence prose description alone (“a square with two disjoint K_4 ’s, one attached at a, c , the other at b, d ”) and found it isomorphic to the graph of [Definition 1.3](#), with the same values $\text{fvs} = 5$, $\text{fp} = 2$. Finally, the brute-force values $\text{cp}(G_1) = 3$ and $\text{cp}(G_2) = 5$ match $2t + 1$, the

85 simple cycles of G_1 were checked against [Lemma 2.3](#) (29 cycles meet each gadget, no two of them disjoint) and [Lemma 2.4](#) (the only gadget-free cycle is C_1), and the node connectivity of G_1 and G_2 was found to be 2.

Remark 3.3 (Consistency with the literature). Since $\text{cp}(G_t) = 2t + 1$ and $\text{fvs}(G_t) = 3t + 2 \leq 2 \text{cp}(G_t)$, the family does not contradict Jones’ Conjecture, and it is not a candidate for doing so. Bärnkopf and Györi [3, Remark 2.5] prove $\text{fvs}(G) \leq 2 \text{fp}(G)$ for every based planar graph; the referee checked that in none of the 32 planar embeddings of G_1 is some face adjacent to every other face, so G_1 is not a based planar graph and there is no conflict.

Remark 3.4 (What is not shown). The construction lives entirely on the 2-cuts $\{p_j, q_j\}$: the force of [Lemma 2.3](#) is that a gadget reachable through only two edges can host at most one packed cycle, and [Lemma 2.6](#) exploits the freedom of embedding across these cuts. G_t is 2-connected but never 3-connected. For 3-connected planar graphs Whitney’s theorem [5] makes the embedding unique up to reflection, so fp is simply the largest set of pairwise vertex-disjoint face boundaries of that embedding, and [Conjecture 1.1](#) is untouched there. The natural repaired question — *does $\text{fvs}(G) \leq 2 \text{fp}(G)$ hold for every 3-connected planar graph G ?* — remains open as far as this note goes. Nor does this note prove any upper bound: [Corollary 1.5](#) shows that no constant $c < 3$ works, not that $c = 3$ does; whether $\text{fvs}(G) \leq 3 \text{fp}(G)$ holds for all planar graphs is open (the known bound $\text{fvs} \leq 3 \text{cp}$ does not imply it, since $\text{fp} \leq \text{cp}$). The ratio 3 is a supremum, not attained: $\text{fvs}(G_t)/\text{fp}(G_t) = 3 - 1/(t + 1) < 3$ for every t . The case “ $t = 0$ ” would be a single K_4 , with $\text{fvs} = 2 = 2 \text{fp}$, so the hypothesis $t \geq 1$ in [Theorem 1.4](#) is needed.

Remark 3.5 (Minimality and novelty). No claim of minimality is made. The referee searched, using the rigorous apex upper bound on fp (so that any hit would be a genuine counterexample), all 1012 planar graphs on at most 7 vertices and all 5 974 connected planar graphs on 8 vertices: no counterexample exists on at most 8 vertices, and on at most 7 vertices the largest ratio fvs / fp is exactly 2, attained by K_4 . (Both fvs and fp are additive over connected components, so disconnected graphs need not be searched separately.) A sweep of the 261 080 connected graphs on 9 vertices did not finish within the referee’s compute budget, so a smallest counterexample has between 9 and 12 vertices. As to novelty, the referee found only three papers citing [1] (among them [3]) and none of them refutes [Conjecture 1.1](#); targeted web searches found no published counterexample. A short observation of this kind could nevertheless be folklore or unindexed, and no priority is claimed.

4 Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The construction and proof were found by the language model gpt-6-astra (reasoning effort max) in a single autonomous pass started on 2026-09-16, as part of a sweep over open problems catalogued from arXiv (catalog id 1912.01570__00, leg attacks_arxiv_astra; original writeup in attacks_arxiv_astra/1912.01570__00/output.md). The model itself claimed the verdict “disproved” with high confidence and disclaimed minimality and literature priority. (2) An independent adversarial referee agent (claude-opus-5) reviewed the writeup on 2026-09-17. It re-derived each of the twelve steps it extracted from the writeup and found all of them valid; checked the conjecture’s wording against both the arXiv and the published EJC text; read [3] and checked consistency with its Remark 2.5; recomputed $\text{fvs}(G_1) = 5$ by exhaustive search and $\text{fp}(G_1) = 2$ by enumerating all 331 776 rotation systems of G_1 ([Remark 3.2](#)); confirmed $\text{fp}(G_t) = t + 1$ for $t \leq 3$ by an independent apex-planarity criterion; rebuilt the graph from prose and checked isomorphism; searched exhaustively for counterexamples on at most 8 vertices; and searched the literature for prior publication. Its verdict was CONFIRMED. Its report is reproduced verbatim in [Appendix B](#). (3) On 2026-09-17 the writeup was rewritten into the present note by claude-fable-5-1. The deviations from the writeup are the following; no mathematical content was changed.

- The text was reorganised: the family G_t is now a definition, the five facts used in the writeup’s

Sections 1–5 are separate lemmas, and full proofs are collected in [Appendix A](#). The value $\text{cp}(G_t) = 2t + 1$, stated with proof in the writeup’s Section 6, was made part of [Theorem 1.4](#).

- The writeup’s Jordan-curve argument (its Section 4) was spelled out as the standalone [Lemma 2.5](#), with the three-face structure of a plane theta graph made explicit and a reference to Diestel [6] for the planar-topology facts used. The writeup’s phrase “when traversing another C_j , use its path $b_j c_j d_j$ ” was corrected to give the actual traversal direction on the descending leg ($d_k c_k b_k$), as the referee noted; this is cosmetic.
- In the proof of [Lemma 2.3](#) the sentence “otherwise it must use both attachment edges” was expanded by the parity argument on the edge cut around V_j .
- The explicit 12-vertex edge list for G_1 , the edge count $13t + 7$ and the statement that G_t is 2-connected but not 3-connected were added (the last from the referee report).
- Citations were added: the published version [2] of the source paper, in which the conjecture is renumbered Conjecture 14 (identified by the referee); Bärnkopf and Györi [3] (named in the problem catalog and read by the referee); Kloks, Lee and Liu [4] for Jones’ Conjecture; Whitney [5]; Diestel [6]. The writeup cited nothing.
- [Remarks 3.1](#) to [3.5](#) were written from the referee report; in particular the scope caveat about 3-connected planar graphs and the interpretation analysis are the referee’s, not the writeup’s.

No human mathematician has checked this argument. We circulate it for human review.

A Full proofs

Throughout, $t \geq 1$ is fixed and $V_j = \{p_j, q_j, r_j, s_j\}$ for $0 \leq j \leq t$.

Lemma 2.1 (Planarity and a face-packing of size $t + 1$). *G_t is planar, and it has a planar embedding in which the $t + 1$ pairwise vertex-disjoint triangles $p_j q_j s_j$ ($0 \leq j \leq t$) all bound faces. Hence $\text{fp}(G_t) \geq t + 1$.*

Proof. Consider first the auxiliary graph A_t on the vertices a_i, b_i, c_i, d_i ($1 \leq i \leq t$) with the edges of the cycles C_i , the chain edges $d_i b_{i+1}$ ($1 \leq i < t$), the chords $a_i c_i$ ($1 \leq i \leq t$) and one closing edge $d_t b_1$. It has a plane drawing: place $b_i = (3i, 0)$, $a_i = (3i + 1, 1)$, $c_i = (3i + 1, -1)$, $d_i = (3i + 2, 0)$, draw the edges of C_i as the sides of the diamond $b_i a_i d_i c_i$, the chord $a_i c_i$ as the vertical segment inside it, each chain edge $d_i b_{i+1}$ as the horizontal segment from $(3i + 2, 0)$ to $(3i + 3, 0)$, and the closing edge $d_t b_1$ as an arc from $(3t + 2, 0)$ to $(3, 0)$ passing above all diamonds (say through $(3t/2 + 5/2, 3)$). These curves meet only at common endpoints, so A_t is planar.

Now let $e_1, \dots, e_t, e_0$ be the chords $a_1 c_1, \dots, a_t c_t$ and the closing edge $d_t b_1$, and let $x_j y_j$ denote e_j , with $(x_j, y_j) = (a_j, c_j)$ for $j \geq 1$ and $(x_0, y_0) = (b_1, d_t)$. Note that G_t is obtained from A_t by deleting each e_j and adding the gadget Q_j together with the edges $x_j p_j$ and $y_j q_j$. Fix j . Choose a point m_j in the interior of the drawn edge e_j and a closed disc D_j centred at m_j so small that D_j meets the drawing of A_t only in a sub-arc of e_j , and so that the discs $D_0, \dots, D_t$ are pairwise disjoint. Inside D_j draw $Q_j \cong K_4$ as a triangle $p_j q_j r_j$ with s_j in its interior joined to the three corners (the standard plane drawing of K_4), placed so that p_j and q_j lie in the interior of D_j and the whole drawing of Q_j is disjoint from e_j . Delete the sub-arc $e_j \cap D_j$ and, within the interior of D_j and outside the triangle $p_j q_j r_j$, join the point where e_j enters D_j from the side of x_j to p_j , and the point where it enters from the side of y_j to q_j , by two disjoint arcs avoiding the drawing of Q_j (possible because p_j and q_j both lie on the outer boundary of the plane K_4 inside D_j and the region of D_j outside that triangle is an open annulus-like set containing the two boundary points). The two new arcs, together with the remaining parts of e_j , are the edges $x_j p_j$ and $y_j q_j$. Doing this for every j produces a plane drawing of G_t .

In this drawing the three regions of D_j bounded by the triangles $p_j q_j s_j$, $q_j r_j s_j$, $p_j r_j s_j$ contain no point of the drawing, since everything added or retained inside D_j lies outside the triangle $p_j q_j r_j$ or on the drawing of Q_j itself. Hence each of them is a face, and in particular $p_j q_j s_j$ bounds a face for every $0 \leq j \leq t$. These $t + 1$ triangles are pairwise vertex-disjoint, so $\text{fp}(G_t) \geq t + 1$. $\square$

Lemma 2.2 (Feedback vertex number). $\text{fvs}(G_t) = 3t + 2$.

Proof. Lower bound. Let S be a feedback vertex set. For each i , S contains a vertex of C_i , since otherwise $C_i \subseteq G_t - S$. For each j , S contains at least two vertices of V_j : any three vertices of K_4 induce a triangle, so if $|S \cap V_j| \leq 1$ then $G_t - S$ contains a triangle. The $2t + 1$ sets $V(C_1), \dots, V(C_t), V_0, \dots, V_t$ are pairwise disjoint, so $|S| \geq t + 2(t + 1) = 3t + 2$.

Upper bound. Let $S = \{a_i : 1 \leq i \leq t\} \cup \{p_j, q_j : 0 \leq j \leq t\}$, of size $3t + 2$. Every attachment edge has an end in S (p_j or q_j), and so does every edge of C_i incident with a_i . The remaining edges of $G_t - S$ are: the edges $b_i c_i, c_i d_i$ ($1 \leq i \leq t$), the chain edges $d_i b_{i+1}$ ($1 \leq i < t$), and the edges $r_j s_j$ ($0 \leq j \leq t$). Together these form the path $b_1 c_1 d_1 b_2 c_2 d_2 \dots b_t c_t d_t$ and $t + 1$ disjoint edges $r_j s_j$, a forest. Hence $\text{fvs}(G_t) \leq 3t + 2$. $\square$

Lemma 2.3 (One cycle per gadget). *Fix $0 \leq j \leq t$. Any two cycles of G_t that both contain a vertex of V_j have a common vertex. Consequently, in any cycle packing of G_t at most one cycle meets V_j .*

Proof. Let Z be a cycle of G_t containing a vertex of V_j . If $V(Z) \subseteq V_j$ then Z is a cycle of K_4 , of length 3 or 4, so $|V(Z) \cap V_j| \geq 3$. Otherwise Z has a vertex outside V_j . Traversing Z once, the edges of Z with exactly one end in V_j are passed each time Z enters or leaves V_j , so their number is even, and it is positive since Z has vertices both in and outside V_j . By Definition 1.3 the only edges of G_t with exactly one end in V_j are the two attachment edges of Q_j , one incident with p_j and the other with q_j . Hence Z contains both attachment edges and therefore both p_j and q_j .

Now let Z, Z' be two cycles meeting V_j . If both are inside Q_j they each contain at least three of the four vertices of V_j , so they share a vertex. If neither is inside Q_j both contain p_j (and q_j). If exactly one, say Z , is inside Q_j , then $V(Z)$ is a subset of V_j of size at least 3 and $\{p_j, q_j\} \subseteq V(Z')$, and a 3-subset of a 4-set meets every 2-subset. In all cases $V(Z) \cap V(Z') \neq \emptyset$, which is the first statement, and the second follows at once. $\square$

Lemma 2.4 (Gadget-free cycles). *The cycles of G_t that contain no vertex of $V_0 \cup \dots \cup V_t$ are exactly $C_1, \dots, C_t$.*

Proof. Let $H = G_t - \bigcup_j V_j$. Deleting the gadgets deletes all attachment edges, so H has vertex set $\bigcup_i V(C_i)$ and edge set consisting of the edges of the C_i and the chain edges $d_i b_{i+1}$ ($1 \leq i < t$). The cycles of G_t avoiding every V_j are exactly the cycles of H . Each chain edge $d_i b_{i+1}$ is a bridge of H : $H - d_i b_{i+1}$ has no edge between $V(C_1) \cup \dots \cup V(C_i)$ and $V(C_{i+1}) \cup \dots \cup V(C_t)$. No cycle contains a bridge, so a cycle of H is a cycle of H minus all chain edges, which is the disjoint union of the C_i ; the only cycles there are the C_i themselves. Conversely each C_i is a cycle of G_t avoiding every V_j . $\square$

Lemma 2.5 (Alternating paths block a face). *Let H be a plane graph containing a cycle C and two paths P (from x to y) and R (from u to v), each with at least one edge, such that P and R are internally disjoint from each other and from C , and x, u, y, v are four distinct vertices appearing in this cyclic order on C . Then no face of H is bounded by C .*

Proof. We use the standard facts about plane graphs collected in [6, Section 4.1]: the Jordan curve theorem, that a connected set disjoint from a plane graph lies in a single face of it, that the faces of a plane graph consisting of a cycle C are exactly the two components of $\mathbb{R}^2 \setminus C$, and that if P is an arc (here: the drawing of a path) joining two points of C and otherwise disjoint from C , then P lies in one component S of $\mathbb{R}^2 \setminus C$, and $S \setminus P$ has exactly two components, whose frontiers are the two cycles of $C \cup P$ through P [6, Lemma 4.1.2]. In particular the plane graph $C \cup P$ has exactly three faces: the side of C not containing P , bounded by C , and the two components of $S \setminus P$, bounded by the cycles $C_1 = xCu yPx$ and $C_2 = yCv xPy$ (writing xCu for the x - y arc of C through u and yCv for the other one). The vertex u lies on C_1 but not on C_2 , and v on C_2 but not on C_1 , since x, u, y, v are distinct and occur in this cyclic order.

Let $\mathring{R}$ denote the drawing of R minus its endpoints u, v . It is connected and disjoint from $C \cup P$, so it lies in one face F of $C \cup P$. Since the drawn edges of R incident with u and v approach these points from within F , both u and v lie on the frontier of F . F is not the face bounded by C_1 (its frontier misses v) nor the one bounded by C_2 (its frontier misses u), so F is the side of C not containing P . Consequently each of the two sides of C contains a drawn edge of H (of P on one side, of R on the other).

Suppose that some face Φ of H were bounded by C . Φ is connected and disjoint from C , so $\Phi \subseteq S'$ for one side S' of C . Φ is open, and it is also relatively closed in S' : a point of S' in the closure of Φ lies on the frontier of Φ or in Φ , and the frontier of Φ is C , which is disjoint from S' . As S' is connected and $\Phi \neq \emptyset$, we get $\Phi = S'$. But S' contains a drawn edge of H , whereas a face contains no point of the drawing, a contradiction. $\square$

Lemma 2.6 (No C_i is facial). *For every $1 \leq i \leq t$ and every planar embedding of G_t , the cycle C_i does not bound a face.*

Proof. Fix i and a planar embedding of G_t . Let $P_i = a_i p_i q_i c_i$; its internal vertices lie in V_i , so it is internally disjoint from C_i . Let R_i be the following b_i - d_i walk: starting at b_i , if $i > 1$ follow the chain edge $b_i d_{i-1}$ and then $d_{i-1} c_{i-1} b_{i-1}$, then $b_{i-1} d_{i-2}$, and so on, until b_1 is reached; then take $b_1 p_0 q_0 d_t$; then, if $i < t$, follow $d_t c_t b_t$, the chain edge $b_t d_{t-1}$, $d_{t-1} c_{t-1} b_{t-1}$, and so on, until reaching d_i via the chain edge $b_{i+1} d_i$. (For $t = 1$, R_1 is simply $b_1 p_0 q_0 d_1$.) The vertices of R_i are b_i, d_i , all vertices of C_k other than a_k for $k \neq i$, and p_0, q_0 ; they are pairwise distinct, so R_i is a path, and its internal vertices avoid $V(C_i) \cup V(P_i)$, since $V(P_i) \setminus V(C_i) \subseteq V_i$ and $V_i \cap V_0 = \emptyset$. Thus P_i and R_i are internally disjoint from each other and from C_i , and their endpoints a_i, b_i, c_i, d_i are distinct and occur in this cyclic order on $C_i = a_i b_i c_i d_i a_i$. Lemma 2.5 (with $x = a_i, u = b_i, y = c_i, v = d_i$) shows that C_i bounds no face of the embedding. $\square$

Theorem 1.4 (Main theorem). *For every integer $t \geq 1$ the graph G_t is a simple planar graph of order $8t + 4$ and maximum degree 4 with*

$$\text{fvs}(G_t) = 3t + 2, \quad \text{fp}(G_t) = t + 1, \quad \text{cp}(G_t) = 2t + 1.$$

Proof. G_t is simple by construction, of order $8t + 4$ and maximum degree 4 (the degree count follows from Definition 1.3: a_i, c_i have degree 3, b_i, d_i have degree 3 because each is incident with exactly one chain or attachment edge, r_j, s_j have degree 3 and p_j, q_j degree 4). Planarity and $\text{fp}(G_t) \geq t + 1$ are Lemma 2.1; $\text{fvs}(G_t) = 3t + 2$ is Lemma 2.2.

$\text{fp}(G_t) \leq t + 1$. Fix a planar embedding and a family $\mathcal{F}$ of pairwise vertex-disjoint cycles each bounding a face of it. By Lemma 2.6 no C_i belongs to $\mathcal{F}$, so by Lemma 2.4 every member of $\mathcal{F}$ contains a vertex of some V_j . Assign to each member one such j . By Lemma 2.3 two disjoint cycles cannot both meet the same V_j , so the assignment is injective and $|\mathcal{F}| \leq t + 1$. Since the embedding was arbitrary, $\text{fp}(G_t) \leq t + 1$.

$\text{cp}(G_t) = 2t + 1$. The cycles $C_1, \dots, C_t$ together with one triangle chosen in each Q_j are $2t + 1$ pairwise vertex-disjoint cycles, so $\text{cp}(G_t) \geq 2t + 1$. Conversely, in a cycle packing at most one cycle meets each V_j (Lemma 2.3), giving at most $t + 1$ cycles meeting some gadget, and the cycles meeting no gadget are among $C_1, \dots, C_t$ (Lemma 2.4), giving at most t more. Hence $\text{cp}(G_t) \leq 2t + 1$. $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-opus-5, 2026-09-17) on the original writeup. Its step numbers refer to that writeup, not to this note. The scripts it mentions are in `verification_astra/scripts/1912.01570__00/`. Treat it as machine output, not as an authority.

B.1 Interpretation

The conjecture under attack is, verbatim from the published version (EJC 28(4), P4.5, 2021, Conjecture 14; identical to Conjecture 2 of arXiv:1912.01570v1):

Conjecture 14. For every planar graph G , $\text{fvs}(G) \leq 2 \cdot \text{fp}(G)$, where $\text{fp}(G)$ is the maximum size of a face-packing of G , i.e., a cycle-packing where, for some embedding of G , every cycle bounds a face.

and the same paper’s Section 1 fixes the other two terms: “A cycle packing of a (multi)graph G is a set of vertex disjoint cycles that appear in G as subgraphs”; “A feedback vertex set of a (multi)graph G is a set S of vertices such that $G - S$ is a forest.”

The writeup uses exactly these definitions: vertex-disjoint cycles, all bounding faces in one common embedding, maximised over embeddings. I checked both the arXiv v1 and the published EJC text: **there is no connectivity, degree, simplicity or 2-connectivity hypothesis** — the quantifier is literally “for every planar graph G ”. The counterexample G_1 is a simple, connected, 2-connected planar graph on 12 vertices with maximum degree 4, so it is inside the stated scope.

Three further points in favour of the interpretation being the intended one:

1. The writeup uses the *most generous* reading of fp — the maximum over **all** embeddings. Any stricter reading (fp of a fixed plane graph, as used e.g. in Definition 1.5 of arXiv:2401.07376) only makes fp smaller and the conjecture harder, so the refutation is robust to the convention. $\text{fvs}(G_1) = 5 > 4 \geq 2 \cdot \text{fp}(G_1)$ under either reading. It is even robust to the *looser-than-intended* reading in which each packed cycle may be facial in an embedding of its own rather than in one common embedding: my computation of the set of cycles that are facial in **some** embedding of G_1 found exactly 20 such cycles, and the largest pairwise vertex-disjoint subfamily of that whole union is still only 2. So no quantifier placement in the definition rescues the conjecture on this graph.
2. The authors’ own sentence immediately before the conjecture — “the example of many nested disjoint cycles shows that the embedding cannot be fixed” — shows they were deliberately quantifying over embeddings and were consciously thinking about low-connectivity / disconnected configurations. So a 2-connected example exploiting 2-cuts is not an off-scope technicality that the authors implicitly excluded.
3. The conjecture is presented as a deliberately provocative strengthening (“To highlight how little we understand around Jones’ Conjecture, we conclude by posing the following stronger conjecture”), i.e. a one-paragraph closing remark rather than a scrutinised central claim. A small counterexample is entirely plausible for such a statement.

There is **no interpretation gaming** here: this is a genuine refutation of the conjecture as written and as published. I would expect the original authors to accept it.

The one legitimate scope reservation (see Caveats) is that the construction leans essentially on 2-cuts, so the *interesting* question — whether $\text{fvs} \leq 2 \cdot \text{fp}$ survives for 3-connected planar graphs, where Whitney makes the embedding essentially unique — is untouched. The writeup does not raise this, but it also does not claim to.

B.2 Step-by-step findings

step	label	note
S1. Construction of G_t : cycles $C_i = a_i b_i c_i d_i$, K_4 gadgets Q_j , edges $a_i p_i$, $c_i q_i$, $d_i b_{i+1}$, $b_1 p_0$, $d_t q_0$; order $8t+4$, simple, max degree 4	VALID	Reproduced verbatim in code. $n = 8t+4$ confirmed for $t = 1..6$ (12, 20, 28, 36, 44, 52); $m = 13t+7$; degree multiset is exactly $\{3(8t), 4(4 \text{ on the } p_j, q_j)\}$; graph simple and connected.
S2. Planarity of G_t (“draw C_i as diamonds in a row, diagonals inside, closing edge outside, then replace each diagonal by a gadget”)	VALID	The replacement argument is correct: p_j and q_j are adjacent in K_4 , hence lie on a common face of any plane K_4 , so the gadget can be substituted inside a small disk around the edge. Independently confirmed: <code>check_planarity</code> returns True for $t = 1..6$.
S3. $\text{fp}(G_t) \geq t+1$ via the embedding with outer triangle $p_j q_j r_j$ and s_j inside, giving disjoint facial triangles $p_j q_j s_j$	VALID	The three internal faces $p q s$, $q r s$, $p r s$ of the K_4 survive the substitution (only the outer triangle’s region merges with the ambient face). Verified concretely: in an explicitly constructed planar embedding I traced the actual face list (Euler $V-E+F=2$ checked each time) and found $t+1$ pairwise vertex-disjoint facial triangles for $t = 1..6$. Stronger remark (not in the writeup, and it makes S3 embedding-independent): in <i>every</i> embedding the gadget’s outer boundary must be a triangle containing both p_j and q_j , so a facial triangle on $\{p_j, q_j, \cdot\}$ always exists. $\text{fvs}(K_4) = 2$ because any 3 vertices of K_4 induce a triangle; the t cycles and $t+1$ gadgets are vertex-disjoint, so the bounds add.
S4. $\text{fvs}(G_t) \geq 3t+2$ (≥ 1 per C_i , ≥ 2 per K_4 , all vertex-disjoint)	VALID	$G_t - S$ is exactly the path $b_1 c_1 d_1 b_2 \dots d_t$ plus $t+1$ isolated edges $r_j s_j$. Verified: <code>is_forest</code> ($G_t - S$) is True for $t = 1, 2$. Brute force over <i>all</i> vertex subsets gives $\text{fvs}(G_1) = 5$ and $\text{fvs}(G_2) = 8$, matching $3t+2$ exactly.
S5. $\text{fvs}(G_t) \leq 3t+2$ via $S = \{a_i\} \cup \{p_j, q_j\}$	VALID	

step	label	note
S6. At most one cycle of a vertex-disjoint packing can meet a given gadget Q_j	VALID	Case analysis is complete and correct (a cycle inside Q_j uses ≥ 3 of 4 vertices; a cycle leaving Q_j must use both attachment edges hence both p_j and q_j ; $3+3 > 4$ and $3+2 > 4$). Brute-forced over all simple cycles: of the 29 ($t=1$) / 149, 109, 109 ($t=2$) cycles meeting a gadget, the number of vertex-disjoint pairs is 0 in every case.
S7. No C_i bounds a face in any planar embedding	VALID	The witness structure is right: $P_i = a_i p_i q_i c_i$ and a $b_i d_i$ path R_i through Q_0 are internally disjoint with endpoints alternating on C_i , so $C_i \cup P_i \cup R_i$ is a K_4 -subdivision and the two paths must lie on opposite sides of the Jordan curve C_i ; hence both sides carry edges and C_i is not facial. I verified the two paths exist and are internally disjoint and avoid C_i internally for $t = 1, 2, 3$. Independently: exhaustive enumeration of all 32 planar embeddings of G_1 shows C_1 is facial in none of them.
S8. After deleting all gadgets, the joining edges are bridges, so the only gadget-free cycles are the C_i	VALID	Brute-forced: the only simple cycles of G_t avoiding every Q_j are exactly $\{a_i b_i c_i d_i\}$, $i = 1..t$ (checked $t = 1, 2$).
S9. $fp(G_t) = t+1$ (every facial cycle meets a gadget by S7+S8; ≤ 1 per gadget by S6; $t+1$ gadgets)	VALID	The decisive check: exhaustive enumeration of all 331 776 rotation systems of G_1 found 32 planar ones, and the maximum number of pairwise vertex-disjoint facial cycles over all of them is 2 . A second, independent method (apex criterion, see below) gives $fp \leq t+1$ for $t = 1, 2, 3$.
S10. $t = 1$: $fvs(G_1) = 5 > 4 = 2 \cdot fp(G_1)$, refuting the conjecture	VALID	Both numbers verified by independent brute force.
S11. No constant $c < 3$ can work, since $fvs(fp) = (3t+2)/(t+1) = 3 - 1/(t+1) \rightarrow 3$	VALID	Arithmetic correct ($t=1$: $5/2 = 2.5 = 3 - 1/2$). $\sup_t (3 - 1/(t+1)) = 3$, so for any $c < 3$ some t violates $fvs \leq c \cdot fp$. The writeup correctly adds that this gives no upper bound, only optimality of 3 <i>if</i> one exists.
S12. $cp(G_t) = 2t+1$, so the family does not contradict Jones' Conjecture itself	VALID	Lower bound: all C_i plus one triangle per gadget. Upper bound: ≤ 1 per gadget (S6) plus $\leq t$ gadget-free cycles (S8). Brute-forced: $cp(G_1) = 3$, $cp(G_2) = 5$, matching $2t+1$. And $fvs = 3t+2 \leq 2(2t+1) = 4t+2$ for all $t \geq 1$, consistent with Jones.

No step is labelled GAP or ERROR. The only imprecisions I found are cosmetic: in S7 the writeup says “when traversing another C_j , use its path $b_j c_j d_j$ ” while the traversal direction is actually $d_j \rightarrow c_j \rightarrow b_j$ on the descending leg (immaterial), and S3’s phrase “its three internal triangular faces remain faces” is correct but would read better as “the outer triangle’s face is destroyed, the three internal ones survive”.

B.3 Reference check

The writeup invokes **no external theorem** — it is fully self-contained (only the Jordan curve argument and $fvs(K_4) = 2$, both elementary). So there is nothing for it to have misquoted. What I checked instead is the target statement itself and the surrounding literature:

- **arXiv:1912.01570v1** (fetched PDF, extracted text). Conjecture 2 reads exactly as the prompt quotes it. Section 1 defines cycle packing as vertex-disjoint cycles and fvs as the minimum feedback vertex set size. **Confirmed.**
- **Published version**, *Jones' Conjecture in subcubic graphs*, Electronic J. Combin. 28(4) P4.5 (2021), fetched from the QUB repository. The conjecture is renumbered **Conjecture 14** and is word-for-word identical, including “For every planar graph G ” with no extra hypotheses. **Confirmed — the arXiv statement was not silently amended in the journal version.**
- **arXiv:2401.07376**, Bärnkopf & Györi, *Jones' conjecture for Halin graphs and a bit more* (the only related work named in the catalog). Fetched and read. Definition 1.5 defines face packing (with a fixed embedding), and **Remark 2.5** states “Every based planar graph G satisfies $fvs(G) \leq 2 \cdot fp(G)$ ”. This is a published theorem that would *contradict* the writeup if G_1 happened to be a based planar graph, so I checked: across all 32 planar embeddings of G_1 , **no face is adjacent to every other face**, so G_1 is not based planar and there is no conflict. **Confirmed consistent.**
- **Priority.** Semantic Scholar lists only 3 papers citing arXiv:1912.01570 (Bärnkopf–Györi 2024; *Far-apart Erdős–Pósa property of long cycles* 2026; *Minimum maximal matchings in cubic graphs* 2020); none refutes Conjecture 14. Several targeted web searches for a counterexample to the face-packing conjecture returned nothing. The catalog’s own 2026-05 auto-review also found nothing. I therefore found **no evidence of prior publication**, but I cannot certify a negative — a short note of this kind could easily be unindexed or folklore. This is the one reason the verdict is CONFIRMED rather than provably novel.

B.4 Computational check

Scripts under `verification_astra/scripts/1912.01570__00/` (Python 3, networkx 3.6.1, nauty geng). All of them run in well under a minute except `minimality.py` 9.

`build.py` — builds G_t and brute-forces the invariants.

```
t=1: n=12 m=20 degrees {3,4} simple planar connectivity=2
      fvs = 5 (= 3t+2) witness {a1,p0,p1,q0,q1}; writeup's S leaves a forest: True
      85 simple cycles; cp = 3 (= 2t+1)
      Q_0: 29 cycles meet it, 0 vertex-disjoint pairs; Q_1: 29 cycles, 0 disjoint pairs
      cycles avoiding every gadget: [a1,b1,c1,d1] (only C_1)
t=2: n=20 m=33 planar; fvs = 8 (= 3t+2); cp = 5 (= 2t+1)
      Q_0/Q_1/Q_2: 149/109/109 cycles meet them, 0 disjoint pairs in each
      cycles avoiding every gadget: C_1 and C_2 only
t=3: n=28 m=46 planar, max degree 4
```

`embeddings.py` — the decisive check. Enumerates **every** rotation system of G_1 (all cyclic orders at every vertex: $6^4 \cdot 2^8 = 331\,776$), keeps those with $V - E + F = 2$ (i.e. genus 0 = sphere embeddings, the most permissive reading of “some embedding”), traces the faces and takes the maximum vertex-disjoint family of facial cycles.

```
enumerated 331776 rotation systems -> 32 planar ones
distinct cycles facial in SOME planar embedding: 20 (all 20 meet a gadget)
*** fp(G_1) = 2 (writeup claims 2) *** witnessing packing: {p1,r1,s1}, {p0,q0,r0}
C_1 facial in some planar embedding: False
```

So $fvs(G_1) = 5 > 4 = 2 \cdot fp(G_1)$, established by two independent brute forces with no appeal to the writeup’s reasoning.

`prose_check.py` — guards against my having mis-transcribed the construction. I rebuilt the $t=1$ graph a second time, from the writeup’s *prose* description only (“take a square $abcd$ and two disjoint K_4 ’s. Connect one K_4 to a,c , and the other to b,d , using distinct attachment vertices in each K_4 ”), with fresh vertex names and no reference to the formal edge list:

```
prose graph: n = 12 m = 20; isomorphic to build(1): True; planar: True
fvs = 5; fp = 2 (again over all 32 planar rotation systems, 20 facial-realizable cycles)
Conjecture 14 requires fvs <= 2*fp, i.e. 5 <= 4: False
```

`sanity_and_t2.py` — validates the enumerator and extends to larger t .

- Enumerator sanity (Whitney: a 3-connected planar graph has exactly 2 rotation systems of genus 0, a mirror pair): $K_4 \rightarrow 2$, cube $Q_3 \rightarrow 2$, octahedron $\rightarrow 2$, $C_5 \rightarrow 1$, $K_5 \rightarrow 0$, $K_{\{3,3\}} \rightarrow 0$. All as expected.
- A second, independent upper bound on fp : if C bounds a face in some embedding then a new apex vertex can be placed in that face and joined to all of $V(C)$, so $G + apex(V(C))$ is planar. Hence $fp(G) \leq \max$ disjoint family among cycles passing this test — a *rigorous* upper bound. On G_1 the apex test selects exactly the same 20 cycles the exhaustive enumeration found to be facial (sets equal), which validates it.

```
t=1: 20 apex-candidate cycles, fp <= 2 (= t+1); no C_i passes the test
t=2: 36 apex-candidate cycles, fp <= 3 (= t+1); no C_i passes the test
t=3: 58 apex-candidate cycles, fp <= 4 (= t+1); no C_i passes the test
```

- The K_4 -subdivision witnesses of S_7 were printed and checked internally disjoint for $t = 1, 2, 3$, e.g. $t=1$: $P = a1-p1-q1-c1$, $R = b1-p0-q0-d1$.

`lower_bound.py` — exhibits a concrete planar embedding of G_t , traces its real face list (Euler $V - E + F = 2$ verified) and reads off the disjoint facial triangles:

```
t=1: F=10, disjoint facial cycles in this embedding: 2 (>= t+1)
t=2: F=15 -> 3      t=3: F=20 -> 4      t=4: F=25 -> 5
t=5: F=30 -> 6      t=6: F=35 -> 7
```

Combining with the apex upper bound: **$fp(G_t) = t+1$ verified exactly for $t = 1, 2, 3$** , and $fp(G_t) \geq t+1$ verified for $t \leq 6$.

`consistency_based.py` — cross-check against the published Remark 2.5 of arXiv:2401.07376: across all 32 planar embeddings of G_1 , the number of embeddings in which some face is adjacent to every other face is **0**, so G_1 is not a based planar graph and the counterexample does not contradict that theorem.

`small_search.py` / `minimality.py` — independent search for counterexamples among small planar graphs, using the rigorous apex upper bound (so any hit would be a genuine counterexample). All 1012 planar graphs on ≤ 7 vertices from the networkx atlas: **0 counterexamples**, and the largest ratio fvs/fp attained is exactly **2.0**, by K_4 ($fvs = 2$, $fp = 1$) — i.e. the conjecture is tight but unbroken there. Exhaustive over all connected graphs on 8 vertices via `geng`: 11 117 connected, 5 974 planar, 1 195 with $fvs \geq 3$, **0 counterexamples**. (Disconnected graphs need not be searched separately: both fvs and fp are additive over connected components, so a disconnected counterexample would force a smaller connected one.) The $n = 9$ sweep (261 080 connected graphs) did not finish inside this review’s compute budget and I stopped it, so I make **no claim** about the minimum order of a counterexample beyond “at least 9 vertices”. The writeup itself explicitly disclaims minimality, so nothing hinges on this.

Everything I attempted to check, I was able to run; nothing load-bearing was skipped.

B.5 Caveats

1. **Only the literal statement falls.** The construction depends critically on the 2-cuts $\{p_j, q_j\}$: the whole force of Section 3 is that a gadget reachable by only two edges can host at most one packed cycle. G_t is 2-connected but never 3-connected (verified: $\text{node_connectivity} = 2$ for $t = 1, 2$). For 3-connected planar graphs Whitney’s theorem makes the embedding unique up to reflection and fp becomes the dual’s independence number, and the conjecture is untouched there. A referee should regard the natural repaired question — *does $\text{fvs} \leq 2 \cdot \text{fp}$ hold for 3-connected planar graphs?* — as still open. The writeup does not mention this, which is the main thing I would ask an author to add.
2. **The ratio-3 claim is an asymptotic supremum, not attained.** $\text{fvs}/\text{fp} = 3 - 1/(t+1) < 3$ for every t ; no member of the family achieves ratio 3. The writeup states this correctly but a careless reader could take “tending to 3” as “reaching 3”.
3. **No upper bound is proved.** The writeup is explicit about this: it shows $c < 3$ is impossible, not that $c = 3$ works. Whether $\text{fvs} \leq 3 \cdot \text{fp}$ holds is open (the known $\text{fvs} \leq 3 \cdot \text{cp}$ does not imply it, since $\text{fp} \leq \text{cp}$ — exactly the point the source paper itself makes).
4. **Minimality is unclaimed and unproven.** 12 vertices is not shown to be least; I verified exhaustively only that no counterexample exists on ≤ 8 vertices. Whether a 9-, 10- or 11-vertex counterexample exists is open as far as this review goes.
5. **Small- t edge cases.** The theorem is stated for $t \geq 1$ and that is necessary: “ $t = 0$ ” would be a lone K_4 with $\text{fvs} = 2$, $\text{fp} = 1$, which satisfies the conjecture with equality. The degenerate inputs the conjecture might be probed with ($k = 0$, forests, empty graph, disconnected graphs) are all irrelevant here since a single explicit connected graph suffices.
6. **Convention robustness.** I checked the refutation under the *generous* reading (max over all sphere embeddings). Under a plane/fixed-outer-face reading, or the fixed-embedding reading of Definition 1.5 in arXiv:2401.07376, fp can only be smaller, so the refutation is a fortiori valid. The one reading under which it would fail — fp counting edge-disjoint rather than vertex-disjoint facial cycles — is excluded by the source paper’s own definition of cycle packing (“vertex disjoint cycles”).
7. **Priority not certifiable.** I found no publication of this counterexample, but absence of evidence in three targeted searches plus a 3-item citation list is weak evidence of absence. The writeup’s own caveat (“no claim of ... literature priority”) is appropriate.

B.6 Referee summary

I set out to break this writeup and could not. The conjecture it attacks is quoted correctly from both the arXiv and the published EJC version, with no hypothesis silently dropped: it really does say “for every planar graph G ”. The 12-vertex graph G_1 — a 4-cycle with a K_4 hung on each of its two opposite-vertex pairs — is a genuine counterexample, and I verified both of its invariants by brute force without using any of the writeup’s reasoning: $\text{fvs}(G_1) = 5$ by exhaustive subset search, and $\text{fp}(G_1) = 2$ by enumerating all 331 776 rotation systems of the graph, isolating the 32 that are planar, and maximising disjoint facial cycles over every one of them. A second, structurally independent method (the apex-planarity criterion) reproduces the identical set of facial-realizable cycles and extends $\text{fp}(G_t) = t+1$ to $t = 2, 3$, with $\text{fp} \geq t+1$ confirmed by explicit traced embeddings up to $t = 6$. Every one of the twelve steps I extracted from the writeup is VALID; the two combinatorial lemmas doing the real work (at most one packed cycle per K_4 gadget; no C_i is facial in any embedding, via a K_4 -subdivision and a Jordan-curve parity argument) are both correct as written and both confirmed by exhaustive computation. The result is also consistent with the published literature: the family has $\text{cp}(G_t) = 2t+1$ so it does not touch Jones’ Conjecture itself, and G_1 is not a based planar graph, so it does not contradict Remark 2.5 of arXiv:2401.07376. The honest reservations are about scope rather than correctness: the construction lives entirely on 2-cuts, so the interesting 3-connected case is untouched, no upper bound replaces the refuted one, minimality is unproven (I checked exhaustively that no counterexample exists on 8 or fewer vertices), and I cannot certify that this counterexample is unpublished, though nothing in the three papers citing the source suggests otherwise. Subject to that last point, the claimed disproof stands.

References

- [1] M. Bonamy, F. Dross, T. Masařík, W. Nadara, Ma. Pilipczuk and Mi. Pilipczuk, Jones’ Conjecture in subcubic graphs, [arXiv:1912.01570](#) (2019), Conjecture 2.
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- [3] P. Bärnkopf and E. Győri, Jones’ conjecture for Halin graphs and a bit more, [arXiv:2401.07376](#) (2024), Definition 1.5 and Remark 2.5.
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- [6] R. Diestel, *Graph Theory*, 5th edition, Graduate Texts in Mathematics 173, Springer, 2017, Section 4.1.